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Agreement between cognitive screening tools: a comparison of MoCA, ACE-III, M-ACE, and MoCA-MIS

Monika McAtarsney-Kovacs^a, Raju Sapkota^a, Shahina Pardhan^a and Ian van der Linde^{a,b,c} 

^aVision & Eye Research Institute, School of Medicine, Anglia Ruskin University, Cambridge, UK; ^bSchool of Computing and Information Science, Anglia Ruskin University, Cambridge, UK; ^cCognition and Neuroscience Group, ARU Centre for Mind and Behaviour, Anglia Ruskin University, Cambridge, UK

ABSTRACT

Introduction: Early detection of cognitive decline is critical, yet commonly used screening tools may yield divergent results. This study examined the alignment of four widely used instruments in a community-dwelling cohort.

Method: Seventy-nine participants ($M_{age} = 71.20$, $SD = 6.29$; 39 male, 40 female) were recruited from the general community and assessed under standardized laboratory conditions. Participants completed the Montreal Cognitive Assessment (MoCA) and the Addenbrooke's Cognitive Examination-III (ACE-III), from which scores for the MoCA Memory Index Score (MoCA-MIS) and the Mini-Addenbrooke's Cognitive Examination (M-ACE) were also calculated. Test agreement was examined using observed agreement and Cohen's kappa, and score relationships were evaluated using Pearson correlations.

Results: All but one test pair (M-ACE vs MoCA-MIS) showed strong score correlations. However, agreement between MoCA and ACE-III for classifying participants as below threshold was only fair. MoCA identified twice as many participants as impaired compared with ACE-III, raising the possibility of either greater sensitivity or higher false-positive classification. Overall, 41% of participants ($n = 32$) scored below threshold on at least one measure, but only 6% ($n = 5$) were below threshold across all four. Of four participants with a prior mild cognitive impairment (MCI) diagnosis, only one, who had more pronounced deficits, scored below threshold in all tests.

Conclusions: Despite strong inter-test correlations, substantial divergence in classification outcome was observed, particularly between MoCA and ACE-III. Reliance on a single screening instrument risks misclassification, with implications for both over- and under-diagnosis. Greater harmonization between screening tools is needed to improve diagnostic consistency. Clinical practice should therefore combine multiple-instrument approaches, potentially including premorbid function, and interpret screening test scores cautiously. Comprehensive neuropsychological assessment may provide a more complete picture, particularly in cases where diagnostic confidence is low, although this is not routine practice in all services.

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Introduction

Cognitive decline, particularly in older adults, can significantly impact quality of life and is often a precursor to more severe conditions such as dementia. Early detection is crucial; cognitive screening tools such as the Montreal Cognitive Assessment (MoCA) and the Addenbrooke's Cognitive Examination-III (ACE-III) are widely used for this purpose.

The MoCA (Nasreddine et al., 2005) is a brief (10–15 minute) screening tool which, compared to earlier tools, is widely held to be highly sensitive to Mild Cognitive Impairment (MCI), a condition characterized by a subtle but noticeable decline in memory and

thinking skills. MoCA was developed to overcome sensitivity limitations and ceiling effects in the Mini-Mental State Examination (MMSE; Folstein et al., 1975), which has subsequently been found to be less accurate than both MoCA and ACE-III (Ciesielska et al., 2016; Giebel & Challis, 2017; Jongsiriyanyong & Limpawattana, 2018; Senda et al., 2020). MoCA assesses multiple cognitive domains, including attention, visuospatial/executive function, naming, memory, language, abstraction, delayed recall, and orientation. The test has a maximum score of 30, with scores 26–30 indicating normal cognition, 18–25 indicating possible MCI, 10–17 indicating

CONTACT Ian van der Linde  ian.vanderlinde@aru.ac.uk  School of Computing and Information Science, Anglia Ruskin University, East Road, Cambridge CB1 1PT, UK

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moderate impairment, and scores below 10 indicating severe impairment.

The sensitivity of MoCA may come at the expense of specificity, leading some to suggest that it may over-identify individuals as cognitively impaired (Luis et al., 2009). To compensate for this, it is recommended that factors such as education, cultural background, and language proficiency are considered when interpreting scores. For instance, a score adjustment of +1 for ≤ 12 years of education has been suggested (Nasreddine et al., 2005).

A supplemental memory-focused metric, the Memory Index Score (MoCA-MIS), can also be calculated, based on performance in the Delayed Recall task. The examiner reads 5 words aloud, for which memory is tested following an ~5-minute interval. All words that are recalled without cues being provided score 3 points, giving a maximum of 15 points. If a word cannot be recalled, first a semantic cue is given (i.e., “it is a body part” for the word “face”). If this cue elicits a correct response, 2 points are awarded. If the semantic cue does not facilitate retrieval, a three-alternative multiple-choice cue is provided. If this elicits a correct response, 1 point is awarded. If a word is not recalled following both cue types, 0 points are awarded. The MoCA-MIS score includes cued responses, but only un-cued recall contributes to the full MoCA score (up to 5 points), so despite the Delayed Recall Task being part of the main test, MoCA-MIS does capture additional information. Although there is no standard guidance for MoCA-MIS score interpretation (for instance, a threshold for normal memory function), Kaur et al. (2018) reported mean scores of 12.2 (SD = 2.8) for normal controls and 7.8 (SD = 3.8) for participants with amnesic MCI, the MCI subtype in which memory is the main cognitive domain affected. Similarly, a MoCA-MIS score < 7 has been suggested as an indication for individuals with MCI to progress to AD (Julayanont et al., 2014).

The ACE-III is more comprehensive than MoCA, taking approximately 15–20 minutes to administer (Hsieh et al., 2013), and has a maximum score of 100. Its more detailed nature may account for reduced sensitivity relative to the MoCA in some contexts (for instance, when the deficit is limited to a smaller number of cognitive domains). A validation study by Hsieh et al. (2013) indicated that the cutoff scores established for the earlier ACE-R (88/100 and 82/100) remain appropriate for identifying performance below the threshold for normal cognition, with 88 being the most widely used and scores below 82 suggestive of more serious impairment. ACE-III also provides sub-domain scores useful for differential diagnosis: attention (/18), memory

(/26), fluency (/14), language (/26) and visuospatial skills (/16). As with MoCA, when interpreting ACE-III test scores, it has been suggested that years of education, age, and premorbid IQ should be considered (Bruno & Schurmann Vignaga, 2019).

An abbreviated variant called M-ACE (mini-ACE) also exists that uses a subset of ACE-III questions (although unlike the MoCA-MIS, it does not capture any information that is not present in the full test). M-ACE has a maximum score of 30 and takes approximately 5 minutes to administer. It has been suggested that those with normal cognition should score ≥ 25 (Hsieh et al., 2015), and that a score < 22 may indicate dementia (Bruno & Schurmann Vignaga, 2019). M-ACE is sometimes used to decide if a full ACE-III should be run, particularly if a near-threshold score is obtained, or if the patient subsequently describes symptoms in cognitive domains not examined in detail in the M-ACE.

While MoCA, MoCA-MIS, ACE-III and M-ACE are all effective screening instruments, they can serve complementary roles. The MoCA is particularly useful for rapid screening of early cognitive impairment, whereas the more comprehensive ACE-III can also be valuable for detailed cognitive profiling and differential diagnosis. The MoCA-MIS can be used to examine memory function more specifically. The M-ACE, a short test often used to decide whether a full ACE-III should be run, can be interpreted independently in some circumstances. Despite this it is common for just one test to be used in a particular clinical setting, highlighting the need for their periodic comparison.

While individual test characteristics are well-documented, their relative performance is a subject of debate. Table 1 summarizes findings from key comparative studies. There is no universal agreement as to the superior test, or even the optimum threshold values for MCI and dementia screening. The words “cut-off” and “threshold” are also used inconsistently, and it is sometimes unclear whether proposed thresholds are intended to be inclusive (e.g., $\leq x$) or exclusive (e.g., $< x$).

Cheung et al. (2015) conducted a study in New Zealand with 84 participants, including 37 diagnosed with mild dementia and 47 controls, and reported that both MoCA and ACE-III performed well in discriminating mild dementia from the control group. Using their published results, we note that ACE-III had a slight advantage in terms of both sensitivity (the proportion of correctly identified true positives) and specificity (the proportion of correctly identified true negatives) individually, and in a combined metric (balanced accuracy, the arithmetic mean of sensitivity and specificity). Using a small cohort ($n = 33$), all with

Table 1. Diagnostic performance and suggested thresholds for MoCA, ACE-III and M-ACE in earlier studies. Balanced accuracy was calculated as the mean of sensitivity and specificity. Where multiple tests are compared, grey highlighting shows the highest metric (sensitivity, specificity, balanced accuracy, AUC).

	Sample Size					Screening Test	MCI				Mild Dementia				Moderate Dementia				CIND			
	N	Healthy	MCI	Dementia	Other		Sensitivity	Specificity	Balanced Acc	AUC	Threshold	Sensitivity	Specificity	Balanced Acc	AUC	Threshold	Sensitivity	Specificity	Balanced Acc	AUC	Threshold	
Nasreddine et al. (2005)	277	90	94	93		MoCA	90.0	87.0	88.5	<26		100.0	87.0	93.5	<26							
Hsieh et al. (2013)	86	25	61			ACE-III	100.0	96.0	98.0	88		93.0	100.0	96.5	82							
Cheung et al. (2015)	84	47	37			MoCA						78.4	83.0	80.7	0.84	≤20						
						ACE-III						81.1	85.1	83.1	0.89	≤76						
Jubb & Evans (2015)	59	6	20	33		ACE-III						79.0	96.0	87.5	0.91	<81						
Matias-Guiu et al. (2017a)	200	25	47	47	81 ^a	ACE-III	76.6	75.0	75.8	0.82	73/74 ^c	94.0	95.0	94.5	0.97	63/64						
Matias-Guiu et al. (2017b)	160	68	92			MoCA						70.7	83.8	77.25	0.86	≤14						
						ACE-III						95.6	67.7	81.7	0.90	≤73						
						M-ACE						82.6	89.7	86.15	0.94	≤15						
Peixoto et al. (2018)	90	30	30	30		MoCA	75.0	53.6	64.3	0.78	24	96.0	81.0	88.5	0.95	21						
						ACE-III	81.3	67.9	74.6	0.82	84	93.8	85.7	89.75	0.98	83						
Wang et al. (2019)	256	136	120			MoCA	86.5	97.7	92.1	0.96	23											
						ACE-III	97.3	90.7	94	0.98	85											
Li et al. (2019)	176	55	64	57		MoCA	67.0	77.0	72	0.76	24/25	71.0	93.0	82	0.91	21/22						
						ACE-III	75.0	89.0	82	0.88	88/89	94.0	83.0	88.5	0.95	74/75						
Senda et al. (2020)	249	50	94	105		MoCA	86.0	64.0	75	0.83	24/25	78.0	77.0	77.5	0.85	20/21						
						ACE-III	80.0	86.0	83.0	0.89	88/89	82.0	86.0	84.0	0.93	75/76						
						M-ACE	77.0	72.0	74.5	0.86	25/26	80.0	85.0	82.5	0.92	18/19						
Valles-Salgado et al. (2024)	140	84	56			MoCA	78.6	65.5	72.05	0.79	22											
						ACE-III	82.1	72.6	77.4	0.86	78											
						M-ACE	78.6	81.0	79.8	0.87	20											
McCarthy et al. (2024)	80		80			ACE-III									0.85	≤76	0.15	≤59				
						M-ACE									0.81	≤19	0.18	≤13				
Grasina et al. (2024)	196	71	125 ^b			M-ACE																

a: 48 with subjective memory complaints, 33 with other neurodegenerative diseases

b: cognitive impairment with no dementia (CIND)

c: discriminating between aMCI and healthy participants with subjective memory complaints

dementia confirmed using clinical tests administered not more than six months prior, Giebel and Challis (2017) assessed the sensitivity of MMSE, MoCA, and ACE-III relative to measures of everyday activity, concluding ACE-III to be superior in identifying the everyday functional impairments that differentiate mild and moderate dementia. Matias-Guiu, Cortés-Martínez, et al. (2017) assessed 200 participants (25 healthy controls, 48 with subjective memory complaints, 47 amnesic MCI, 47 mild AD and 33 with other neurodegenerative disease) and found ACE-III to be an effective screening tool for detecting early-stage AD vs normal, with slightly lower performance in discriminating MCI vs early-stage AD. In another study from the same authors, the diagnostic accuracy of ACE-III, M-ACE (extracted from ACE-III post-hoc), Memory Impairment Screen (MIS), Mini-Mental State Examination (MMSE), MoCA, and RUDAS (Rowland Universal Dementia Assessment Scale; Storey et al., 2004) for AD screening were compared (Matias-Guiu, Valles-Salgado et al., 2017). Using 160 participants, 92 diagnosed with mild AD and 68 healthy controls, among all tools examined, M-ACE was found to yield the highest specificity and balanced accuracy in mild AD detection, while ACE-III had higher sensitivity.

Peixoto et al. (2018) found the Portuguese version of the ACE-III to have better specificity and sensitivity overall compared to the MoCA when differentiating healthy controls, MCI, and mild dementia in a sample of 90 participants (30 controls, 30 MCI, and 30 with dementia), except that sensitivity was found to be higher in detecting mild dementia with MoCA. Li et al. (2019) examined the Chinese version of ACE-III for the detection of MCI and mild dementia, comparing it to the MMSE and MoCA. Using 176 participants (64 with MCI, 57 with mild dementia, and 55 controls) they found ACE-III to yield the higher sensitivity and specificity in MCI detection. Using the values reported, ACE-III is also seen to yield higher balanced accuracy than MoCA in mild dementia detection, although greater specificity for MoCA was reported.

Wang et al. (2019) compared the Chinese versions of MoCA and ACE-III with 256 participants (120 with MCI and 136 healthy controls) and report no significant difference in diagnostic performance, with greater sensitivity for ACE-III but greater specificity for MoCA (although calculating balanced accuracy indicates a slight advantage for ACE-III). Senda et al. (2020) assessed 249 participants (50 controls, 94 with MCI, and 105 with dementia) with a range of screening tools including ACE-III, M-ACE, and MoCA. They report superior performance for ACE-III and M-ACE in both

MCI and dementia diagnosis, although the values reported also indicate greater sensitivity for MoCA in MCI detection. Valles-Salgado et al. (2024) compared tests including MoCA, ACE-III, and M-ACE with 140 participants (84 controls, 56 MCI). ACE-III and M-ACE were found to have higher specificity and sensitivity than MoCA for MCI detection. They suggest that M-ACE offers a good balance between performance and administration time. Calculating balanced accuracy also shows M-ACE to perform best overall, despite marginally higher sensitivity for ACE-III. A study by Grasina et al. (2024) with 196 participants (71 controls, and 125 with cognitive impairment in long-term care), also found M-ACE, in its Portuguese translation, to be a valid and reliable screening tool to discriminate between controls and those with cognitive impairment with no dementia (CIND).

Hsieh et al. (2013) concluded that the ACE-III is a good screening tool to detect AD using 86 participants (61 with dementia and 25 healthy controls). Jubb and Evans (2015) assessed the diagnostic accuracy of ACE-III for dementia detection using 59 participants and proposed a lower score threshold of < 81 , suggesting that this provided improved diagnostic performance, although reported sensitivity is lower than the value report in Hsieh et al. (2013). McCarthy et al. (2024) also proposed lower threshold scores for ACE-III and M-ACE for mild dementia detection (ACE-III ≤ 76 ; M-ACE ≤ 19) using 80 participants with dementia, observing slightly better performance, expressed as AUC, for ACE-III. Performance for moderate dementia was very low with both tests, with surprisingly poor AUC scores.

Several meta-analyses have compared these screening instruments. Larner and Mitchell (2014) examined the accuracy of the ACE, ACE-R, and MMSE in detecting dementia. Examining numerous studies with varying sample sizes and population types, it was concluded that ACE-R (the second revision of this test) had superior performance in dementia detection. A meta-analysis by Tsoi et al. (2015) suggested that, besides the MMSE, the MoCA was best at detecting MCI, and that ACE-R and the Mini-Cog (Borson et al., 2000) were best for dementia detection from among eleven screening tests compared. In contrast, a meta-analysis by Ciesielska et al. (2016) compared studies using the MoCA and MMSE for MCI detection in those aged over 60, suggesting MoCA to be superior in MCI screening. A systematic review by Abd Razak et al. (2019) found the MoCA for MCI and ACE-III for dementia to be the most accurate screening tools in a primary care setting. Davis et al. (2015), in a Cochrane review of the MoCA, reported high sensitivity (0.94) but modest specificity

(0.60), concluding that its use cannot be recommended as the primary diagnostic tool on the available evidence. They highlighted the need for studies that avoid case-control designs and instead apply diagnostic tests and reference standards prospectively. In a subsequent Cochrane review, Beishon et al. (2019) highlight a lack of sufficient high-quality data in primary care settings to support the use of ACE-III or M-ACE as the primary tool for dementia or MCI screening, and advocate for the use of full clinical evaluation.

Using the metric of balanced accuracy (calculated by us), a holistic measure that takes the average of specificity and sensitivity, many of the above studies found ACE-III to be superior in detecting mild dementia (Cheung et al., 2015; Li et al., 2019; McCarthy et al., 2024; Peixoto et al., 2018; Senda et al., 2020), with one study giving superior performance for the abbreviated M-ACE (Matías-Guiú, Valles-Salgado, et al., 2017). Even when authors suggest that tests elicit broadly the same performance, numerically, we find a similar outcome for MCI, with balanced accuracy being higher for ACE-III (Peixoto et al., 2018; Li et al., 2019; Senda et al., 2020; Wang et al., 2019), with a different study finding superior performance for the M-ACE (Valles-Salgado et al., 2024). Despite this, some reviews have suggested superior performance for the MoCA (Abd Razak et al., 2019; Tsoi et al., 2015). The story is less clear when we look at sensitivity and specificity separately, where MoCA has been found to yield greater values than ACE-III in some cases. For mild dementia, Peixoto et al. (2018) report higher sensitivity for MoCA vs ACE-III, and Li et al. (2019) report higher specificity. For MCI, Wang et al. (2019) report higher specificity for MoCA vs ACE-III, and Senda et al. (2020) report higher sensitivity.

All studies summarized in Table 1 recruited patients through memory clinics, neurology departments, or used clinical databases of patients with existing diagnoses, except one study that included participants from long term care facilities (Grasina et al., 2024). These clinical studies include those who have already contacted health services for assessment, at which point their cognitive decline is likely to be more advanced, but do not include those at an earlier stage, resulting in a potentially non-representative sample. Unlike prior studies which relied on clinical samples, often with confirmed diagnoses and possibly more advanced cognitive decline, our community-based sample may better represent individuals at earlier stages, including those with undiagnosed MCI, at whom these screening tests are really directed. Specifically, in existing studies, if a participant scored below the threshold of

normal cognition without a diagnosis they were considered a “false positive.” In contrast, in our study, if someone scored below the threshold without a previous formal diagnosis of MCI, given the prevalence of undiagnosed MCI in the community (Davis et al., 2015; Liu et al., 2024), rather than assuming diagnostic error, we remained agnostic as to the validity of this outcome.

Administering these tests to the same participant group enables their diagnostic utility to be evaluated, for instance enabling their degree of agreement (or lack thereof) to be observed, which may contribute to helping clinicians determine which instrument is more appropriate (Carson et al., 2012), or whether multiple instruments should be used. In this study, we administered MoCA and ACE-III (including calculating sub-scores MoCA-MIS and M-ACE) to a cohort of older volunteers in which a surprising 41% exhibited below-threshold cognition in at least one test using standard thresholds, rather than revised thresholds proposed by some of the studies summarized in Table 1.

The novelty of this study lies in its use of a community-based sample, and its combined use of correlation and agreement metrics to examine test concordance among four screening tools. Our primary aim was to examine consistency between MoCA and ACE-III in a community-dwelling sample, not to estimate diagnostic accuracy against a “gold standard.” While in the U.S.A., brief screeners such as the MoCA and ACE-III are often supplemented by a full neuropsychological assessment, which may include instruments such as the Clinical Dementia Rating (CDR; Morris, 1993), in the UK and elsewhere they, together with self-report and informant history, typically serve as the primary tools for initial dementia diagnosis following referral by a general practitioner (GP; equivalent to a primary care physician). This practice is consistent with NHS-implemented NICE Guideline NG97 (National Institute for Health and Care Excellence, 2018). This underscores the importance, particularly in countries like the UK, of ensuring that these screening measures are well aligned (as they are sometimes used interchangeably) and are judiciously interpreted.

Materials and methods

Participants

A total of 79 older participants were recruited, mean age = 71.20 (SD = 6.29), age range 53–82, 39 male and 40

Table 2. Scores and descriptive statistics for each test

Test	Score min, max (range)	Score mean (SD)	Number sub-threshold (%)
MoCA (raw)	13, 30 (17)	26.76 (2.50)	16 (20.25%)
MoCA (education-adjusted)	13, 30 (17)	26.91 (2.52)	16 (20.25%)
MoCA-MIS	0, 15 (15)	12.24 (2.97)	24 (30.38%)
ACE-III	52, 100 (48)	93.89 (6.53)	8 (10.13%)
M-ACE	5, 30 (25)	27.00 (3.62)	11 (13.92%)

female. All participants had English as their first language to help ensure that test instructions were understandable. The mean years of education was 16.75 (SD 3.51). University ethics approval was obtained, and the study followed the guidelines of the Declaration of Helsinki. Participants were recruited from the Join Dementia Research (JDR) volunteer database managed by the NIHR (National Institute of Health Research) [$n = 64$], the University of The Third Age (U3A) [$n = 6$], advertising by posters and flyers displayed at the university campus [$n = 7$], and a local retirement home [$n = 2$]. All participants had normal or corrected-to-normal vision and hearing. Exclusion criteria included the following self-reported conditions: low vision, known history of neurological or psychological disorders (excluding MCI), brain injury, epilepsy, dyslexia, stroke, degenerative diseases such as Parkinson's, depression, anxiety and chronic alcohol and drug use. Four participants presented with a preexisting clinical diagnosis of MCI. These participants, although small in number, were included so that we did not actively exclude cases near the decision boundary or reduce ecological validity.

Procedure

Participants completed the MoCA version 8.1 (Nasreddine et al., 2005), and ACE-III (Hsieh et al., 2013) using British English translations, where available. MoCA was administered by a certified rater (ID GBMCAMO710560679-01). Tests were completed in random order in a quiet, well-lit lab without distractions. All data were collected during a single visit in an unhurried manner, and participants were reminded regularly that they may take rest breaks when desired. Demographic data collected included age and years of formal education, which was calculated from the highest qualification obtained. To ensure that sensory issues did not affect test performance, brief vision and hearing assessments were conducted. Normal vision, with correction if needed, was assessed using MNREAD Acuity Chart 2 (Mansfield & Legge, 1990), with all participants achieving a minimum visual acuity of LogMAR 0.4. Hearing ability, with assistive devices if required, was

evaluated by having participants listen to five prerecorded sound clips of color names (e.g., “blue,” “black,” “red”) and repeat each word aloud. All participants repeated all five words correctly. No participants were excluded due to impaired vision or hearing.

Analysis

Descriptive statistics (mean and SD) were calculated for each test (MoCA, MIS, ACE-III, and M-ACE). Scores were interpreted using established criteria, a cutoff of 88 applied for the ACE-III. The numbers of participants for whom test score interpretations were in agreement (e.g., scores all indicative of normal cognition, or otherwise) vs discrepant (i.e., some scores indicative of normal cognition with others not) were examined. Test agreement was evaluated as observed agreement and Cohen's kappa (Ben-David, 2008; McHugh, 2012). Inter-score relationships were also evaluated using Pearson product moment correlation.

Results

Scores and descriptive statistics for all tests and participants are summarized in Table 2. The mean raw MoCA score was 26.76 (SD 2.50). Thirteen (13) participants had ≤ 12 years of education, so a score adjustment of + 1 was applied (Nasreddine et al., 2005), however this did not have the effect of pushing any participant above the normal cognition threshold (i.e., from a score of 25 to 26). Following this adjustment, the mean MoCA score was 26.91 (SD 2.52). From here, education-adjusted MoCA scores are used only.

Sixteen (16) participants had a MoCA score < 26 (the normal cognition threshold). Fifteen (15) had a score in the range 18–25 (indicating MCI), and one (1) had a score in the range 10–17 (indicating moderate impairment). Of the four (4) participants with a preexisting diagnosis of MCI, three (3) were correctly identified by MoCA as falling below the threshold for normal cognition (score < 26), with scores of 13, 24, and 25. One (1) participant with a diagnosis of MCI obtained a score in the normal cognition range (27). This means that, of the 16 participants that did not reach the normal cognition

threshold (20.25% of our sample of older adults), 13 did not have a preexisting MCI diagnosis. These participants were referred to their GP for follow-up testing.

The mean MoCA-MIS score was 12.24 (SD = 2.97) across the entire group. Twenty-four (24) participants had a MoCA-MIS score < 12, below the mean of individuals with normal cognition reported in Kaur et al. (2018). Three (3) of these participants scored < 7, indicating potential conversion to AD (Julayanont et al., 2014). Twelve (12) participants who scored < 12 in MoCA-MIS scored in the normal range in MoCA (> 25). Of the 4 participants with an MCI diagnosis, two scored in the normal range (≥ 12) and two scored < 12, indicative of possible amnesic MCI based on the findings of Kaur et al. (2018), from which one scored < 7, indicating possible transition to AD based on the results of Julayanont et al. (2014).

Across all participants, the mean ACE-III score was 93.89 (SD = 6.53). Eight (8) participants had an ACE-III score < 88 (the threshold for normal cognition, indicating possible MCI or more severe impairment), which is half as many as those falling below the MoCA threshold (10.13% of our sample of older adults, vs 20.25% for MoCA). Of these eight (8), one (1) had a score < 61 (indicative of dementia). Of the four (4) participants with a preexisting diagnosis of MCI, two scored below the threshold for normal cognition (with scores of 52 and 86) and two obtained scores in the normal cognition range (90 and 96). This means that of the eight (8) participants that failed to reach the normal cognition threshold, six (6) did not have a preexisting MCI diagnosis and were also referred to their GP. Of the two participants with a preexisting MCI diagnosis that fell below the ACE-III normal cognition threshold, only one (1) also fell below the normal cognition threshold for MoCA. A fuller evaluation of test agreement/disagreement is provided later. Had the more liberal threshold for normal cognition of 82/100 been used for ACE-III (Hsieh et al., 2013), only two (2) participants would have fallen below the threshold (2.53% of our sample), and this would have excluded three of the four participants with an existing MCI diagnosis.

The mean M-ACE score was 27.00 (SD = 3.62) across all participants. Eleven (11) participants scored below the normal cognition threshold for M-ACE (i.e., < 25)

and three (3) of these participants scored in the normal range for both the ACE-III (≥ 88) and MoCA (≥ 26), three (3) scored under the normal cognition threshold for the ACE-III (< 88) with a MoCA score in the normal range (≥ 26). Five (5) participants scored outside the normal range in all three tests (MoCA, ACE-III, and M-ACE). Three (3) participants scored < 21 (indicating dementia) while their MoCA score was in the normal range (≥ 26). To probe these discrepancies, pairwise occurrences of agreement/disagreement were examined.

Across MoCA, MIS, ACE-III, and M-ACE, 32 participants (41%) fell below the normal cognition threshold in at least one test (union), whilst only five (6%) participants fell below the normal threshold in all four tests (intersection). Considering only the four scores (MoCA, MIS, ACE-III, and M-ACE), there are six possible pairwise comparisons in terms of whether the normal cognition threshold was met.

Remaining agnostic as to whether participants were truly below the normal cognition threshold, the degree of agreement between the tests was calculated by dividing the sum counts in which classifications agreed (YY + NN) by the total number of observations ($n = 79$), yielding observed agreement (P_o). This value was then used to calculate Cohen's kappa (κ), which evaluates the degree to which agreement exceeds that which would be expected by chance. These values are presented in Table 3, with interpretations suggested by Landis and Koch (1977).

As one might expect, given that M-ACE is a subscore of ACE-III that includes several ACE-III questions, the largest observed agreement ($P_o = .96$) is between ACE-III and M-ACE, with the same principle also yielding very good agreement between MoCA and MoCA-MIS ($P_o = .80$), perhaps slightly lower reflecting that MoCA-MIS can contain some information not present in MoCA full score. However, overall scores for the full tests (MoCA and ACE-III) also show good observed agreement at $P_o = .82$, despite twice as many participants being identified as below-threshold by MoCA compared to ACE-III. However, interpreting Cohen's kappa (κ), gives a slightly less favorable assessment of the relationship between MoCA and ACE-III, with only a "fair" agreement ($\kappa = .33$), and, as one might expect, the best agreement being found between tests

Table 3. Observed agreement (P_o) and Cohen's kappa (κ) for each test pairing.

		YY	YN	NY	NN	P_o	κ	Interpretation
MoCA	ACE-III	60	3	11	5	.82	.33	fair
MoCA	M-ACE	57	6	11	5	.78	.25	fair
MoCA	MoCA-MIS	51	12	4	12	.80	.47	moderate
ACE-III	M-ACE	68	3	0	8	.96	.82	almost perfect
ACE-III	MoCA-MIS	53	18	2	6	.75	.26	fair
M-ACE	MoCA-MIS	51	17	4	7	.73	.26	fair

and their own sub-scores (with “almost perfect” agreement between ACE-III and M-ACE, and “moderate” agreement between MoCA and MoCA-MIS).

Although it is highly likely that several participants without a prior diagnosis of MCI in fact did have MCI, summarizing results (above) for the small subset who had a formal diagnosis ($n = 4$) is useful to determine how many were correctly detected as sub-threshold by each test. Of the four, $n = 3$ were identified as sub-threshold by MoCA, $n = 3$ by MoCA-MIS, $n = 2$ by ACE-III, and $n = 3$ by M-ACE.

It is also important to note the number of participants that scored at threshold, 1 point below or above the cutoff point, as these borderline cases may lead to classification errors. Here, 34 participants were

borderline for the MoCA (scores 25–27), 5 were borderline for ACE-III (scores 87–89), 16 for M-ACE (scores 24–26), and 30 for MoCA-MIS (scores 11–13). While the more compressed score range M-ACE, MoCA, and MoCA-MIS means the number of borderline cases must be interpreted cautiously, it does show the potentially large impact of corrections such as the education adjustment for MoCA (Nasreddine et al., 2005) described earlier, and the vulnerability of shorter tests to inattention, mishearing, or other potential sources of error.

Across all participants, the correlation between MoCA and ACE-III scores was $r = .70$, $p < .001$, which is classified as “large” (Cohen, 1988). A scatterplot showing the correlation between these scores, with a least-squares line of best fit, is shown in Figure 1.

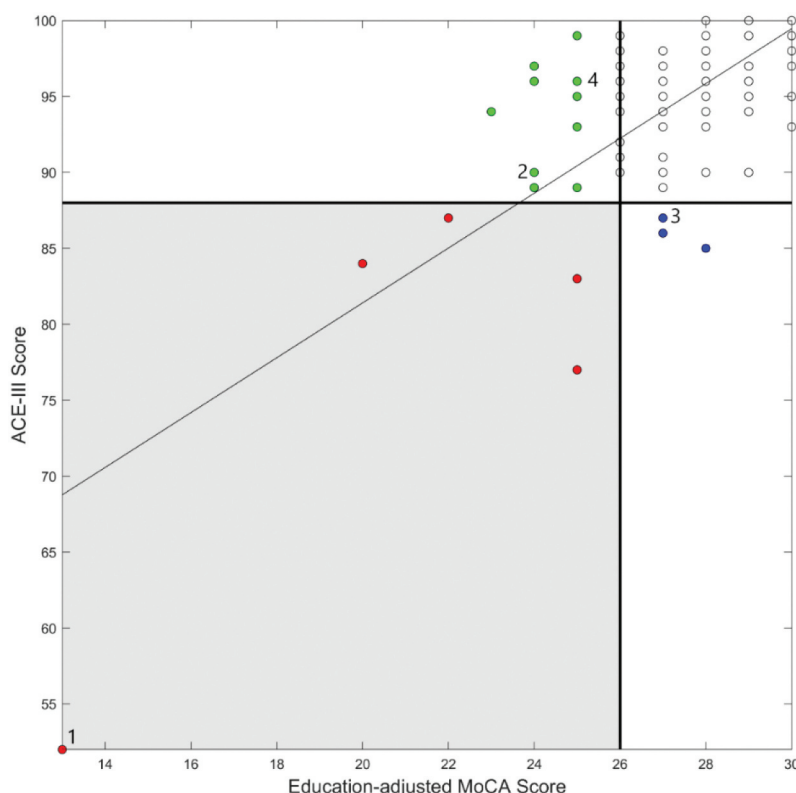


Figure 1. Scatterplot of MoCA and ACE-III scores with “normal cognition” thresholds shown as vertical (MoCA) and horizontal (ACE-III) bars, with intersection shown as shaded region. Red indicates participants who fell below normal cognition threshold for MoCA and ACE-III. Green indicates participants who fell below the normal cognition threshold for MoCA but not ACE-III. Blue indicated participants who fell below the normal cognition threshold for ACE-III but not MoCA. Participants with an existing diagnosis of MCI are highlighted by numbers 1-4.

Table 4. Pairwise correlations between tests

		<i>r</i>	<i>p</i>	Interpretation
MoCA	ACE-III	.79	<.001	large
MoCA	M-ACE	.62	<.001	large
MoCA	MoCA-MIS	.82	<.001	large
ACE-III	M-ACE	.94	<.001	large
ACE-III	MoCA-MIS	.55	<.001	large
M-ACE	MoCA-MIS	.50	<.001	medium

Only 5 participants scored below the “normal cognition” thresholds in *both* MoCA and ACE-III (red points), with most discrepant results showing an ACE-III “pass” and a MoCA “fail” ($n = 11$, green points [note: two points overlap, so only 10 are visible]), with fewer in the converse group (MoCA “pass” and ACE-III “fail,” $n = 3$, blue points). Other pairwise correlations are summarized in Table 4 with interpretations suggested by Cohen (1988).

Supplementary Material A shows a correlation matrix in which all test scores, sub-scores, and demographics variables are correlated. MoCA does not have official section sub-scores like ACE-III, therefore these have been calculated based on the cognitive domain highlighted in each question label. Notably, neither age nor education were found to correlate significantly with MoCA, MoCA-MIS, ACE-III, or M-ACE. The ACE-III Memory score positively correlated with MoCA-MIS ($r = .53$, $p < .001$), as one might expect given their focus on the same cognitive domain, while the MoCA Delayed Recall Task showed a weaker association $r = .39$, $p < .001$.

Our sample scored slightly lower across most of the tests compared to available normative data. The mean MoCA total score (26.91) is slightly lower than the normative data reported by Kenny et al. (2013) of 27.2 (SD = 2.2) for aged 70 with primary or no education, 28.3 (SD = 1.8) with secondary education, and 29.0 (SD = 1.3) with tertiary or higher education. Data provided by Kenny et al. (2013) is based on representative sample of community-living population of Ireland without severe cognitive impairment. Comparing with normative data provide by Sachs et al. (2022) our sample scored above the 50th percentile across all levels of education in both white males and females, except white females with graduate degree, in this category our data is only above the 25th percentile. Mean MoCA scores were not calculated separately for female participants with degree-level education in our study. This normative data (Sachs et al., 2022) was provided by a cognitively normal sample of diverse older U.S. adults.

Our mean full ACE-III score (93.89) is lower than the mean total score 95.4 (SD = 3.3) of the healthy controls provided by Hsieh et al. (2013), although the mean age 66.1. (SD = 7.4) of their control group was lower compared to our sample. The mean full ACE-III score is also lower than the normative data from The University of Sydney (2019), which reported 96 (SD = 2.7) for controls with the mean age of 68.7 (SD = 7), or data provided by So et al. (2018) for

healthy controls in which the mean ACE-III score was 95.7 (SD = 3.3) with participants of mean age 69.6 (SD = 6.7). Additionally, normative data provided by The University of Sydney (2019) for age > 65 for the M-ACE is 27.8 (SD = 1.8), slightly higher than was found in our sample (27.00). Our sample with only 4 participants with a pre-diagnosis of MCI had scored lower in MoCA, ACE-III, and M-ACE compared to normative data. This supports the view that our sample may have contained more participants with cognitive impairments than those with a prior diagnosis.

The mean MoCA-MIS score in our study of 12.24 (SD = 2.97) was similar to the control group mean score of 12.2 (SD = 2.8) reported by Kaur et al. (2018), which also had a similar mean age 72.7 (SD = 10).

Discussion

Our findings highlight a critical distinction between correlation and diagnostic agreement across cognitive screening tools. Despite strong score correlations, agreement was often weaker, particularly between the MoCA and ACE-III, where only “fair” concordance was observed. This discrepancy illustrates the danger of assuming that these instruments are interchangeable.

The prevalence of mild cognitive impairment (MCI) in individuals aged ≥ 65 has been estimated at 4–19%, depending on how MCI is defined and interpreted (Livingston et al., 2017). In our sample, three of 68 participants aged ≥ 65 had an existing MCI diagnosis (4%), aligning with the lower bound of this range. If the upper bound (19%) were applied, our sample could potentially have included an additional 10 undiagnosed cases. Despite this, across all four measures (MoCA, MIS, ACE-III, and M-ACE) 41% of our participants fell below the normal cognition threshold in at least one test, which is considerably higher than would be expected by population prevalence alone. These participants would be classified as sub-threshold if just one test were used in their cognitive assessment.

The accurate identification of MCI and dementia are critical, as misclassification can lead to significant psychosocial consequences for patients and caregivers (Mirza et al., 2017; Munawar et al., 2023); false positives may cause unnecessary anxiety and reduce quality of life (Carter et al., 2023; Milby et al., 2017), while false negatives delay access to support and early intervention.

Our study used participants from the general community, whereas earlier studies (e.g., Cheung et al., 2015; Matias-Guiu, Cortés-Martínez, et al. 2017; Matias-Guiu, Valles-Salgado, et al. 2017 Wang et al., 2019) used clinical samples, except one that assessed participants in long-term care (Grasina et al., 2024),

making direct comparison difficult. It is unknown how many of our participants without prior diagnosis were falsely identified by these tests as having impaired cognition due to other factors (e.g., fatigue, inattention, cognitive profile, or other reasons). Alternatively, some of these individuals may genuinely have met the clinical criteria for MCI had they been assessed more fully, including an informant-supported report of subjective cognitive complaints, objective cognitive impairment in one or more domains relative to age and education, intact daily activities, and the absence of dementia (Petersen, 2004).

Although illustrative rather than inferential, in our data, two participants with preexisting MCI diagnoses scored below the normal threshold on MoCA (scores 24 and 25) while obtaining very high scores in the ACE-III (scores 90 and 96). It may be that MoCA was used to make the clinical judgment, which was shared with these individuals, but that this conclusion may not have been reached had ACE-III rather than MoCA been used.

These results illustrate the importance of a clinical assessment that takes into consideration demographic variables and subjective experience, such as scores in the Clinical Dementia Rating (CDR, Morris, 1993) or the Multifactorial Memory Questionnaire (MMQ, Troyer & Rich, 2002), in addition to raw test scores. This will mitigate the observation that test scores may disagree, that some tests may be more liberal than others, and that some tests focus on cognitive domains that may be more/less affected in the individual case or may reflect the presence of differences in existing cognitive profile. This suggests that premorbid measures of baseline cognitive ability may also be useful in the constellation of evidence used to diagnose MCI/dementia, such as reading tests (Bright & van der Linde, 2020; van der Linde & Bright, 2024), enabling the clinician to differentiate between long-standing individual differences and emerging cognitive decline. Nevertheless, the liberal tendency of MoCA could be advantageous in primary care settings where the cost of missing early decline is high, but less so in contexts where false positives carry significant psychosocial consequences.

The large correlation between the MoCA and ACE-III is in line with earlier research (Giebel & Challis, 2017; Wang et al., 2019). Despite this a discrepancy between MoCA and ACE-III is apparent, with twice as many participants falling below the normal cognition threshold in MoCA compared to ACE-III. This is consistent with earlier studies (e.g., Tsoi et al., 2015) proposing that MoCA has greater sensitivity than ACE-III to detect MCI, although potentially at the expense of specificity, with greater numbers of false positives

identified (Luis et al., 2009). There are a number of possible explanations for this. First, the MoCA places greater emphasis on executive function and attention, whereas the ACE-III assesses a broader range of domains. A lower score on the MoCA relative to the ACE-III might suggest specific difficulties in executive functioning or attention. Second, there may be differences in test sensitivity, such that each test may be more sensitive to certain types of cognitive impairment, where the discrepancy could indicate that one test is detecting subtle deficits that the other is not. Third, differences in test conditions, fatigue, or familiarity with test formats may contribute, although in this case it is the shorter test that elicited a greater number of “fails.”

The difference between these tests was particularly striking for the small number of participants that presented with a prior MCI diagnosis. Of four, only one scored under the threshold in both ACE-III and MoCA and had much lower scores across all tests, indicative of dementia, suggesting agreement between MoCA and ACE-III perhaps in more severe cases only. However, this is not an ideal situation, given that an important role of these screening tools is early detection.

In a comparison with MoCA, it has been suggested that the M-ACE is an equally suitable screening tool to screen for MCI (Larner, 2017); however, we observed some disagreement between these tests. For instance, 10 participants scored below the normal threshold in the MoCA whilst obtaining a M-ACE score in the normal range. Conversely, 6 participants scored in the normal range in the MoCA, but fell below the normal threshold for the M-ACE. What is more interesting, although M-ACE scores were derived from the ACE-III scores, 3 participants scored in the normal range for ACE-III but scored below the normal threshold for M-ACE.

In addition to the large correlation between the MoCA and ACE-III, a follow-up analysis examined the association between specific question domains (Supplementary Material A), indicating that the highest correlations with the MoCA score came from the ACE-III attention and language sub-scores. These are the sub-domains that contain more than one identical or almost identical items with the MoCA, such as assessment date and location, the Serial Sevens test (also known as the Descending Subtraction Task), picture naming, and repeating sentences. This may be responsible for the stronger association observed.

Although previous studies have raised concerns about the influence of age and education on cognitive screening results (Bruno & Schurmann Vignaga, 2019; Malek-Ahmadi et al., 2015), our findings showed no significant correlation between either factor and total or sub-domain scores on any test. This suggests that

these measures are relatively independent of age and education, which is beneficial in that it reduces the risk of misclassifying healthy aging or lower educational attainment as cognitive impairment, an issue previously noted in relation to the MoCA (Khaw et al., 2021; O’Caoimh et al., 2016) and ACE-III (Jubb & Evans, 2015; Matias-Guiu et al., 2017; Bruno & Schurmann Vignaga, 2019).

MoCA-MIS may be a better measure of memory function compared to the MoCA Delayed Recall Task alone, and the MoCA-MIS showed a stronger positive association with the ACE-III memory sub-domain score, which is a more comprehensive assessment of the memory abilities compared to the MoCA Delayed Recall Task. This is in line with the findings by Kim et al. (2021), in which a large correlation between MoCA-MIS and SNSB (Seoul Neuropsychological Screening Battery) memory domain scores was reported for MCI participants. The MoCA-MIS may help differentiate whether a memory deficit stems from retrieval or encoding difficulties. An inability to exploit provided cues may suggest impaired memory encoding. In our sample, 12 participants had MoCA scores within the normal range coupled with low MoCA-MIS scores. These individuals may have been compensating for memory difficulties using alternative strategies. However, a small number fell below the normal threshold on the M-ACE ($n = 1$), or on both the M-ACE and ACE-III ($n = 1$). Poor free recall with improved performance with cues is typical in MCI, in which cueing supports retrieval. In contrast, failure to improve even with cues suggests a deficit in encoding or storage, which is more characteristic of AD (Clément et al., 2010; Hudon et al., 2011).

An observation related to the correlation matrix in Supplementary Material A is that MoCA scores correlate more highly with memory-related measures (Delayed Recall Task and MoCA-MIS) than other measures, suggesting that memory is the most important cognitive domain in determining the overall score. This is despite the Delayed Recall Task only contributing 17% of the final MoCA score, and that only the un-cued recall score from MoCA-MIS is used. Similarly, aside from M-ACE, ACE-III also correlated most highly with ACE Memory sub-score than other cognitive domains, despite contributing the same number of points overall as language (26).

Whilst many studies rely on correlation or ROC analysis to compare tests, we found only “fair” concordance between MoCA and ACE-III with Cohen’s kappa. This suggests that despite high correlation, the tests may lead to different clinical outcomes in nearly half of the cases, underscoring the importance of not relying on single test scores for diagnosis. Giebel and Challis

(2017) also used Cohen’s kappa and found “poor” to “fair” agreement between ACE-III and MoCA to differentiate between mild and moderate dementia (only), depending on the threshold used, suggesting that limited agreement between screeners extends beyond MCI to mild and moderate dementia.

This study has several limitations. First, our participant cohort, which although similar in size to Hsieh et al. (2013), Peixoto et al. (2018), McCarthy et al. (2024), Grasiņa et al. (2024), and Cheung et al. (2015), and larger than Giebel and Challis (2017) and Jubb and Evans (2015), was relatively small, and being a community-based sample, had relatively few participants with a prior diagnosis of MCI. However, MCI is highly underdiagnosed in the community with around 92% of expected cases being undiagnosed (Liu et al., 2024). The sample also had relatively high educational attainment. Second, M-ACE scores were derived from ACE-III scores, and this may have decreased the diagnostic accuracy of M-ACE scoring as thresholds may have been established based on independent administration. Third, that there was no objective criterion, such as a full neuropsychological assessment. However, as described in the Introduction, in the UK (and elsewhere), unlike the U.S.A., initial diagnosis following GP referral relies upon MoCA, ACE-III, or other validated measure listed in the NICE guidelines, and a “gold standard” neuropsychological assessment is not typically used. However, future work should evaluate these findings using clinical groups (*viz.*, equal sized groups of participants with MCI and dementia diagnoses) in addition to a sample confirmed to be neurologically healthy through a comprehensive neuropsychological assessment.

Currently, no National Institute for Health and Care Excellence (NICE) guidance for assessing MCI in the UK exists (Dunne et al., 2021). In a surveillance proposal (National Institute for Health and Care Excellence [NICE], 2023) considering recommendations for MCI and extending the existing guidance on dementia (NICE, 2018), it was concluded that MCI is a complex and heterogeneous syndrome and not all MCI cases are related to dementia. Furthermore, research has yet to provide robust evidence to develop MCI guidelines, with the lack of quality publications specifically highlighted in this report. In the absence of NICE guidance, this study contributes evidence on the community-prevalence of those not reaching the normal threshold in MoCA, MoCA-MIS, ACE-III, and M-ACE, and the degree of test concordance.

Conclusion

Our community-based sample likely better represents individuals at earlier stages of cognitive decline,

including undiagnosed MCI, compared to prior studies that use participants with confirmed diagnoses for comparison against participants confirmed to be neurologically healthy. Unlike previous work that classified below-threshold scores without a diagnosis as “false positives,” we remained agnostic, reflecting the prevalence of undiagnosed MCI. Administering MoCA, ACE-III, MoCA-MIS, and M-ACE to the same participants allowed us to assess diagnostic agreement using standard thresholds, complemented by inter-rater reliability and correlation metrics, providing novel insights into test concordance.

Notably, a large proportion of participants scored below threshold in at least one test, exceeding normative expectations. These findings highlight the complexity of defining normal cognition in community samples and underscore the importance of considering demographics, subjective reports, and premorbid function alongside screening scores. Our findings are particularly relevant in countries like the UK, in which screening tests are frequently used for initial dementia diagnosis, in conjunction with self-report and informant history, rather than a fuller diagnostic battery. As demand for effective cognitive assessment grows, including digital tools (Modarres et al., 2023), careful evaluation of existing instruments is crucial to optimize the design of future screening approaches.

Author contributions

Conceptualization: IV, MMK, SP, RS; Methodology: IV, MMK, SP, RS; Software: IV; Validation: IV and MMK; Formal Analysis: IV and MMK; Investigation: IV, SP, RS and MMK; Resources: IV, MMK, RS and SP; Data Curation: IV and MMK; Writing – Original Draft Preparation: IV and MMK; Writing – Review and Editing: IV, MMK, RS and SP; Visualization: IV; Supervision: IV, RS and SP; Project Administration: IV, RS and SP; Funding Acquisition: SP.

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ORCID

Ian van der Linde  <http://orcid.org/0000-0002-8131-5906>

Institutional review board statement

The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Ethics Committee of Anglia Ruskin University (ethics approval code CIS201221, date of approval 4 January 2022).

Informed consent statement

Informed consent was obtained from all subjects involved in the study.

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